

## Encoding $\llbracket \cdot \rrbracket : \text{CCS-VP} \rightarrow \text{CCS}$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n. \llbracket P\{^n/_x\} \rrbracket$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m. \llbracket P \rrbracket \quad e \downarrow m$$

$$\llbracket \tau.P \rrbracket = \tau. \llbracket P \rrbracket$$

$$\llbracket P \mid Q \rrbracket = \llbracket P \rrbracket \mid \llbracket Q \rrbracket$$

$$\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$$

$$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$$

$$\llbracket P[f] \rrbracket = \llbracket P \rrbracket[f'] \quad f'(a_m) = f(a)_m$$

$$\llbracket \text{if } b \text{ then } P \rrbracket = \begin{cases} \llbracket P \rrbracket & b \text{ true} \\ 0 & \text{otherwise} \end{cases}$$

$$\llbracket K(e_1, \dots, e_h) \rrbracket = K_{n_1, \dots, n_h} \quad e_i \downarrow n_i$$

$$K_{n_1, \dots, n_h} \stackrel{\text{def}}{=} \llbracket P\{^{n_1}/_{x_1}, \dots, ^{n_h}/_{x_h}\} \rrbracket$$

$$\text{where } K(x_1, \dots, x_h) \stackrel{\text{def}}{=} P$$

## CCS Operational Semantics

$$\begin{array}{c}
 \frac{}{\alpha.P \xrightarrow{\alpha} P} \text{ACT} \\
 \\
 \frac{P_j \xrightarrow{\alpha} P'_j}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{SUM} \quad j \in I \\
 \\
 \frac{P \xrightarrow{\alpha} P'}{P \mid Q \xrightarrow{\alpha} P' \mid Q} \text{PAR-L} \\
 \\
 \frac{Q \xrightarrow{\alpha} Q'}{P \mid Q \xrightarrow{\alpha} P \mid Q'} \text{PAR-R} \\
 \\
 \frac{P \xrightarrow{\alpha} P' \quad Q \xrightarrow{\tilde{\alpha}} Q'}{P \mid Q \xrightarrow{\tau} P' \mid Q'} \text{COM} \\
 \\
 \frac{P \xrightarrow{\alpha} P'}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{RES} \quad \alpha, \tilde{\alpha} \notin L \\
 \\
 \frac{P \xrightarrow{\alpha} P'}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{REN} \\
 \\
 \frac{P \xrightarrow{\alpha} P'}{K \xrightarrow{\alpha} P'} \text{CONST} \quad K \stackrel{\text{def}}{=} P
 \end{array}$$

For all  $P$  in CCS-VP

$$\begin{aligned} \text{(i)} \quad & P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \\ \text{(ii)} \quad & \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q \end{aligned}$$

$$\text{where } \hat{\alpha} = \begin{cases} a_m & \alpha = a(m) \\ \bar{a}_m & \alpha = \bar{a}(m) \\ \tau & \alpha = \tau \end{cases}$$

Both implications are established by *rule induction*, or induction on the depth of the derivation tree, which is the natural choice here because the transition relation is itself defined inductively by inference rules. The natural alternative would be *structural induction on  $P$* , case-splitting on its syntactic form rather than on the last rule fired. For most forms this coincides with rule induction, but it fails cleanly at **CONST**: a constant  $K$  is a syntactic atom with no sub-term, so structural induction provides no smaller process to apply the IH to. Rule induction avoids this by measuring derivation depth, which strictly decreases at every premise, including the unfolding step inside **CONST**.

For (i) we proceed by induction on the derivation of  $P \xrightarrow{\alpha} P'$ , with a case analysis on the last rule applied. The *base cases* are the three axiom forms of CCS-VP (input, output, and  $\tau$  prefix): for each, the encoding  $\llbracket P \rrbracket$  is a concrete CCS term that admits the required transition directly via the Act or Sum rule. For the *inductive cases* (Sum, Par-L, Par-R, Com, Res, Ren, Const, Cond), the induction hypothesis supplies transitions for the shorter sub-derivation(s), and the matching CCS rule then closes the goal.

For (ii) we instead proceed by induction on the derivation of  $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q$ , with a case analysis on the syntactic form of  $P$ . For each form of  $P$ , the structure of  $\llbracket P \rrbracket$  uniquely determines which CCS rule could have fired last; the induction hypothesis then recovers a CCS-VP witness  $P'$  with  $P \xrightarrow{\alpha} P'$ , and the encoding equations confirm  $\llbracket P' \rrbracket = Q$ .

Throughout both directions the key invariant is that  $\hat{\alpha}$  correctly bridges CCS-VP labels and CCS channels  $a(m) \leftrightarrow a_m$ ,  $\bar{a}(m) \leftrightarrow \bar{a}_m$ ,  $\tau \leftrightarrow \tau$  ensuring that complementary actions in CCS-VP correspond to complementary channels in CCS, so that synchronization is preserved by the encoding.

$$P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket$$

**Base cases**

**Input**  $a(x).P \xrightarrow{a(m)} P\{^m/_x\}$ :

$$\frac{\frac{a_m \cdot \llbracket P\{^m/_x\} \rrbracket \xrightarrow{a_m} \llbracket P\{^m/_x\} \rrbracket}{\sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/_x\} \rrbracket \xrightarrow{a_m} \llbracket P\{^m/_x\} \rrbracket} \text{Sum}}{\llbracket a(x).P \rrbracket \xrightarrow{a_m} \llbracket P\{^m/_x\} \rrbracket} =$$

**Output**  $\bar{a}(e).P \xrightarrow{\bar{a}(m)} P, e \downarrow m$ :

$$\frac{\frac{}{\bar{a}_m \cdot \llbracket P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket} \text{Act}}{\llbracket \bar{a}(e).P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket} =$$

**Silent**  $\tau.P \xrightarrow{\tau} P$ :

$$\frac{\frac{}{\tau \cdot \llbracket P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket} \text{Act}}{\llbracket \tau.P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket} =$$

**Inductive cases**

$$\frac{P_i \xrightarrow{\alpha} P'_i}{\llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_i \rrbracket} \text{IH}$$

**Sum**  $\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j \quad j \in I$ :

$$\frac{\frac{\frac{P_j \xrightarrow{\alpha} P'_j}{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{IH}}{\sum_{i \in I} \llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{Sum}}{\llbracket \sum_{i \in I} P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} =$$

**Par-L**:  $P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2$

$$\begin{array}{c}
\frac{P_1 \xrightarrow{\alpha} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket} \text{IH} \\
\frac{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket}{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket} \text{Par-L} \\
= \\
\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \mid P_2 \rrbracket
\end{array}$$

**Par-R:**  $P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2$

$$\begin{array}{c}
\frac{P_2 \xrightarrow{\alpha} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_2 \rrbracket} \text{IH} \\
\frac{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \mid P'_2 \rrbracket} \text{Par-R} \\
= \\
\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \mid P'_2 \rrbracket
\end{array}$$

**Com**  $P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2$ :

$$\begin{array}{c}
\frac{P_1 \xrightarrow{a(m)} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_1 \rrbracket} \text{IH} \quad \frac{P_2 \xrightarrow{\bar{a}(m)} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_2 \rrbracket} \text{IH} \\
\frac{\llbracket P_1 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_1 \rrbracket \quad \llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_2 \rrbracket}{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \rrbracket \mid \llbracket P'_2 \rrbracket} \text{Com} \\
= \\
\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \mid P'_2 \rrbracket
\end{array}$$

**Res**  $P \setminus L \xrightarrow{\alpha} P' \setminus L, \alpha \notin L \cup \bar{L}$ :

$$\begin{array}{c}
\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH} \\
\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket}{\llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\} \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}} \text{Res} \\
= \\
\llbracket P \setminus L \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \setminus L \rrbracket
\end{array}$$

**Ren**  $P[f] \xrightarrow{f(\alpha)} P'[f]$ , where  $f'(a_m) = f(a)_m$  and  $\widehat{f(\alpha)} = f'(\hat{\alpha})$ :

$$\begin{array}{c}
\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH} \\
\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket}{\llbracket P \rrbracket[f'] \xrightarrow{f'(\hat{\alpha})} \llbracket P' \rrbracket[f']} \text{Ren} \\
= \\
\llbracket P[f] \rrbracket \xrightarrow{\widehat{f(\alpha)}} \llbracket P'[f] \rrbracket
\end{array}$$

**Const**  $K(e_1, \dots, e_h) \xrightarrow{\alpha} R, e_i \downarrow n_i$ :

$$\begin{array}{c}
\frac{P\{n_1/x_1, \dots\} \xrightarrow{\alpha} R}{K_{n_1, \dots} \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket} \text{IH} \\
= \\
\llbracket K(e_1, \dots, e_h) \rrbracket \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket
\end{array}$$

**Cond** if  $b$  then  $P \xrightarrow{\alpha} P'$ ,  $b$  true:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\llbracket \text{if } b \text{ then } P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket}} =$$

$$\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q$$

**Base cases**

**Input** [  $a(x).P$  ]

$$\frac{}{a(x).P \xrightarrow{a(m)} P\{^m/_x\}} \text{Act}$$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/_x\} \rrbracket.$$

The only possible transition fires branch  $a_m$  for some  $m \in \mathbb{N}$ , so  $Q = \llbracket P\{^m/_x\} \rrbracket$ .

Taking  $P' = P\{^m/_x\}$  witnesses  $P \xrightarrow{a(m)} P'$  and  $\llbracket P' \rrbracket = Q$ .

**Output** [  $\bar{a}(e).P$  ]

$$\frac{e \downarrow m}{\bar{a}(e).P \xrightarrow{\bar{a}(m)} P} \text{Act}$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m \cdot \llbracket P \rrbracket \text{ since } e \downarrow m.$$

The unique transition via Act forces  $Q = \llbracket P \rrbracket$ .

Taking  $P' = P$  witnesses  $P \xrightarrow{\bar{a}(m)} P'$  and  $\llbracket P' \rrbracket = Q$ .

**Silent** [  $\tau.P$  ]

$$\frac{}{\tau.P \xrightarrow{\tau} P} \text{Act}$$

$$\llbracket \tau.P \rrbracket = \tau \cdot \llbracket P \rrbracket.$$

The unique transition via Act forces  $Q = \llbracket P \rrbracket$ .

Taking  $P' = P$  witnesses  $P \xrightarrow{\tau} P'$  and  $\llbracket P' \rrbracket = Q$ .

**Inductive cases**

$$\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q} \text{IH}$$

$$\text{Sum} \left[ \sum_{i \in I} P_i \right]$$

$$\frac{\frac{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} Q}{P_j \xrightarrow{\alpha} P'_j \wedge \llbracket P'_j \rrbracket = Q} \text{IH}}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{Sum}$$

Since  $\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$ , any transition must originate from some branch  $j \in I$ .

The IH recovers  $P_j \xrightarrow{\alpha} P'_j$  with  $\llbracket P'_j \rrbracket = Q$ .

Taking  $P' = P'_j$  witnesses  $P \xrightarrow{\alpha} P'$  and  $\llbracket P' \rrbracket = Q$ .

**Par** [  $P_1 \mid P_2$  ]

$\llbracket P_1 \mid P_2 \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket$ .

Any transition of  $\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket$  must have arisen from one of three CCS rules.

- PAR-L

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} Q_1}{P_1 \xrightarrow{\alpha} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2} \text{Par-L}$$

The transition on the left component gives  $Q = Q_1 \mid \llbracket P_2 \rrbracket$ .

The IH recovers  $P_1 \xrightarrow{\alpha} P'_1$  with  $\llbracket P'_1 \rrbracket = Q_1$ .

Taking  $P' = P'_1 \mid P_2$  witnesses  $P \xrightarrow{\alpha} P'$  and  $\llbracket P' \rrbracket = \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket = Q_1 \mid \llbracket P_2 \rrbracket = Q$ .

- PAR-R

$$\frac{\frac{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} Q_2}{P_2 \xrightarrow{\alpha} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2} \text{Par-R}$$

The transition on the right component gives  $Q = \llbracket P_1 \rrbracket \mid Q_2$ .

The IH recovers  $P_2 \xrightarrow{\alpha} P'_2$  with  $\llbracket P'_2 \rrbracket = Q_2$ .

Taking  $P' = P_1 \mid P'_2$  witnesses  $P \xrightarrow{\alpha} P'$  and  $\llbracket P' \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket = \llbracket P_1 \rrbracket \mid Q_2 = Q$ .

- COM

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{a_m} Q_1}{P_1 \xrightarrow{a(m)} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH} \quad \frac{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} Q_2}{P_2 \xrightarrow{\bar{a}(m)} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2} \text{Com}$$

Complementary transitions  $a_m$  and  $\bar{a}_m$  map back to  $a(m)$  and  $\bar{a}(m)$ .

The IH on each side recovers  $P_1 \xrightarrow{a(m)} P'_1$  with  $\llbracket P'_1 \rrbracket = Q_1$  and  $P_2 \xrightarrow{\bar{a}(m)} P'_2$  with  $\llbracket P'_2 \rrbracket = Q_2$ .

Taking  $P' = P'_1 \mid P'_2$  witnesses  $P \xrightarrow{\tau} P'$  and  $\llbracket P' \rrbracket = Q_1 \mid Q_2 = Q$ .

**Res** [  $P \setminus L$  ]

$$\frac{\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q_0} \text{IH}}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{Res } (\alpha \notin L \cup \bar{L})$$

$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$ , so the transition decomposes as  $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0$  with  $Q = Q_0 \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$ .

The transition being unblocked forces  $\hat{\alpha} \notin \{a_m \mid a \in L\}$ , hence  $\alpha \notin L \cup \bar{L}$ .

The IH recovers  $P \xrightarrow{\alpha} P'$  with  $\llbracket P' \rrbracket = Q_0$ .

Taking  $P' \setminus L$  as the witness gives  $P \setminus L \xrightarrow{\alpha} P' \setminus L$  and  $\llbracket P' \setminus L \rrbracket = \llbracket P' \rrbracket \setminus \{a_m \mid a \in L\} = Q_0 \setminus \{a_m \mid a \in L\} = Q$ .

**Ren** [  $P[f]$  ]

$$\frac{\frac{\frac{[[P]] \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge [[P']] = Q_0} \text{IH}}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{Ren}$$

$[[P[f]]] = [[P]][f']$ , so the transition decomposes as  $[[P]] \xrightarrow{\hat{\alpha}} Q_0$  with  $Q = Q_0[f']$ .

The IH recovers  $P \xrightarrow{\alpha} P'$  with  $[[P']] = Q_0$ .

Taking  $P'[f]$  as the witness gives  $P[f] \xrightarrow{f(\alpha)} P'[f]$  and  $[[P'[f]]] = [[P']][f'] = Q_0[f'] = Q$ .

**Cond** [ if  $b$  then  $P$  ]

$$\frac{b \text{ true} \quad \frac{[[P]] \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge [[P']] = Q} \text{IH}}{\text{if } b \text{ then } P \xrightarrow{\alpha} P'} \text{Cond}$$

$[[\text{if } b \text{ then } P]] = [[P]]$  when  $b$  is true.

The transition is directly on  $[[P]]$ .

The IH recovers  $P \xrightarrow{\alpha} P'$  with  $[[P']] = Q$ .

The IH witness  $P'$  directly satisfies  $P \xrightarrow{\alpha} P'$  and  $[[P']] = Q$ .

**Const** [  $K(e_1, \dots, e_h)$  ]

$$\frac{e_1 \downarrow n_1, \dots, e_h \downarrow n_h \quad \frac{[[P\{^{n_1}/_{x_1}, \dots\}]] \xrightarrow{\hat{\alpha}} Q}{P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R \wedge [[R]] = Q} \text{IH}}{K(e_1, \dots, e_h) \xrightarrow{\alpha} R} \text{Const}$$

$[[K(e_1, \dots, e_h)]] = K_{n_1, \dots} \stackrel{\text{def}}{=} [[P\{^{n_i}/_{x_i}\}]]$ .

The transition is on that encoding.

The IH recovers  $P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R$  with  $[[R]] = Q$ .

Taking  $P' = R$  witnesses  $K(e_1, \dots, e_h) \xrightarrow{\alpha} P'$  and  $[[P']] = Q$ .

**Remark (implementation).** The interpreter developed for this project encodes input actions over a bounded interval  $[n_{\min}, n_{\max}] \subset \mathbb{N}$  rather than an infinite sum over all of  $\mathbb{N}$ . Concretely,  $\llbracket a(x).P \rrbracket = \sum_{n=n_{\min}}^{n_{\max}} a_n \cdot \llbracket P\{^n/_x\} \rrbracket$  instead of  $\sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/_x\} \rrbracket$ . The proof above remains structurally identical: the **Input** base case still applies the Sum rule and the  $=$  step, and every inductive case goes through unchanged. The only difference is that operational correspondence holds under the assumption that all communicated values fall within the chosen interval; values outside it have no corresponding branch in the encoding and therefore no matching CCS transition.